

MATH 2123

QUESTION BANK ANALYSIS

Topic-wise Set Distribution System

Past-paper pattern audit • 2015-2025 • 3-out-of-4 strategy

Purpose

Turn the full Math 2123 syllabus into eight predictable preparation sets: four under Section A and four under Section B. Within each set, the most repeated question families are placed first so you can prepare strategically for the requirement to answer any 3 sets out of 4.

MADE BY 2223033

Prepared from uploaded KUET term questions, lecture notes, solved topic collections and reference books.

Exam-first • Repetition-first • Decision-focused

How to Read This Analysis

Important finding

The historical papers do not keep one topic permanently attached to one section or one question number. In several years the Matrix/Laplace/Vector blocks were swapped between Section A and Section B. Therefore, this bank uses a current 2025-style eight-set map, but ranks questions using the full historical repetition pattern.

Statistical scope

- Primary exam audit: KUET Math 2123 term-final papers from 2015 through 2025 in the uploaded “Math 2123 Term Questions” file.
- 2020 is a special shortened 120-mark / 1.5-hour paper with only three sets per section; it is used as supporting evidence, not as the baseline for the normal 4+4 set structure.
- Supporting validation: topic-wise 2015-2024 compilations for theory/proofs, vector integration, beam deflection and unit step functions, plus the uploaded lecture notes/reference books.
- Priority labels are based on recurrence, stability of the question family, and usefulness under the “answer 3 out of 4” rule.

Recommended preparation order

Section	Tier 1	Tier 2	Backup	Suggested default
Section A	Set 1	Set 2 + Set 3	Set 4	Prepare Sets 1-3 first
Section B	Set 5 + Set 7	Set 8	Set 6	Choose 5-7-8 or 5-6-7

Eight-set map at a glance

Set	Dominant topic	Repeat strength	What to expect
A1	Matrix Fundamentals & Core Theorems	VERY HIGH	Definitions + symmetric/skew + basic proofs
A2	Inverse, Rank, Consistency & Dependence	VERY HIGH	Inverse by ERO + rank tests + systems
A3	Echelon/Normal Form & Eigenvalue Block	VERY HIGH	REF/RREF/normal form + eigenvalues/eigenvectors
A4	Vector Kinematics / Work + Cayley-Hamilton	MED-HIGH	Space-curve mechanics + work + CH theorem
B5	Laplace Fundamentals & Direct Transform	VERY HIGH	Existence + properties + periodic/piecewise/unit step
B6	Inverse Laplace, Convolution & ODE/Beam	VERY HIGH	ILT + convolution + differential-equation applications
B7	Vector Differential Calculus	VERY HIGH	grad/div/curl + directional derivative + potential
B8	Vector Integration	VERY HIGH	line + surface + volume integral / flux / work

SECTION A

Recommended core: Sets 1, 2 and 3. Set 4 is the flexible / mixed backup set.

Why these three?

Matrix fundamentals, inverse/rank/consistency, and normal-form/eigenvalue questions recur with remarkable regularity across the decade. Even when sections are swapped, these question families remain among the most dependable high-mark areas.

Set	Topic family	Priority	Typical marks	Risk
Set 1	Matrix fundamentals & proofs	★★★★★	25-40	Low
Set 2	Inverse + rank + linear systems	★★★★★	25-40	Low
Set 3	Normal forms + eigenvalues	★★★★★	25-40	Low
Set 4	Kinematics/work + Cayley-Hamilton	★★★★☆	25-40	Medium

Matrix Fundamentals & Core Theorems

The highest-yield theory-heavy matrix set. Definitions often supply quick, reliable marks.

Historical footprint

Appears repeatedly throughout the matrix-focused papers; especially strong in 2015, 2016, 2017, 2018, 2021, 2023, 2024 and 2025.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Define matrix types with examples	A near-annual question family; orthogonal is especially recurrent (2016, 2018, 2019, 2022, 2023, 2024 plus 2025-type definitions).
2	Express A as symmetric + skew-symmetric	Directly recurring in 2016/2017/2018 and again reflected in 2025.
3	Orthogonal matrix test / prove orthogonality	Repeated in 2016, 2018, 2019, 2022, 2023, 2024.
4	Skew-symmetric properties / diagonal entries are zero	Dedicated proof in 2018; concept repeats in definitions.
5	Idempotent / involutory / nilpotent property question	2018 involutory $\rightarrow$ idempotent; 2025 nilpotent proof; common rotating proof family.
6	Trace / Hermitian / elementary / equivalent definitions	Frequent short-theory subquestions across 2015-2024.

Core preparation checklist

- Prepare concise definitions + one example for: orthogonal, skew-symmetric, singular/non-singular, diagonal, scalar, Hermitian, idempotent, nilpotent, elementary, equivalent, lower/upper triangular, trace.
- Memorize decomposition formulas: $P = (A + A^T)/2$ and $Q = (A - A^T)/2$, with proof of symmetry/skew-symmetry.
- Know matrix-operation properties and common counterexamples ($AB \neq BA$, cancellation pitfalls, etc.).
- Practice small “find parameters so A is symmetric/orthogonal” problems.

Secondary / rotating questions

- Polynomial in a matrix / evaluate a matrix expression.
- Find A and B from linear matrix equations.
- Basic transpose identities and symmetric inverse theorem.

Exam tactic

Bank the short theory marks first. In a 3-of-4 strategy, Set 1 is one of the safest Section A choices because even when the computational part changes, the definitions/proof component stays highly reusable.

Inverse, Rank, Consistency & Linear Dependence

A calculation-heavy set built around row operations and rank logic.

Historical footprint

Inverse-by-elementary-transformation and consistency/rank questions recur across most standard papers from 2015 onward; 2023-2025 show this block particularly clearly.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Inverse by elementary row transformations	Seen repeatedly: 2015, 2017, 2018, 2019 and strongly again in 2024-2025.
2	Consistency of a parameterized linear system	One of the most persistent decade-long question families: parameter value for unique/no/infinite solutions.
3	Rank of a matrix / parameter-dependent rank	Explicit in 2019, 2023, 2024; often embedded in normal-form and system questions.
4	Trivial vs non-trivial solution of homogeneous systems	2015/2016 core theory and revisited through rank-based questions.
5	Linear dependence / independence of vectors	Appears in 2015, 2017, 2021, 2023-2025 in either direct or application form.
6	Condition for inverse to exist	Explicit 2016 and 2025-type treatment; naturally paired with rank/determinant.

Core preparation checklist

- Practice Gauss-Jordan inverse using $[A|I] \rightarrow [I|A^{-1}]$.
- Know invertibility tests: $\det(A) \neq 0$, full rank, unique solution of $AX=B$.
- Use $\text{rank}(A)$ and $\text{rank}([A|B])$ to classify: no solution / unique / infinitely many solutions.
- For homogeneous systems, connect rank, number of variables and non-trivial solutions.
- Practice linear dependence both by row reduction and by finding a non-trivial linear combination.

Secondary / rotating questions

- Network-flow / network-equation setup.
- Solve linear systems by elementary operations or matrix method.
- Adjoint/inverse relation and products involving A^{-1} .

Exam tactic

Be able to do one clean row-reduction workflow that simultaneously gives rank, consistency and inverse logic. This set rewards procedural accuracy more than memorization.

Echelon / Canonical / Normal Form + Eigenvalue Block

Two of the most stable long-form matrix questions are combined here.

Historical footprint

Normal-form/rank reduction appears repeatedly from 2015 through 2025. Eigenvalues/eigenvectors are also a near-annual matrix topic, often with “largest eigenvalue”, inverse-eigenvalue or diagonalization variants.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Echelon → canonical → normal form and rank	Repeated in 2015, 2016, 2017, 2018, 2022, 2023, 2025.
2	Eigenvalues and corresponding eigenvectors	Repeated across 2015-2019 and 2021-2025 in direct or applied form.
3	Largest eigenvalue / eigenvalues of inverse	2015, 2022 and similar variants.
4	Find P,Q for normal form PAQ	2017/2018/2022-style recurring transformation problem.
5	Diagonalization / $P^{-1}AP$	2015 and 2019-style questions; high-value backup.
6	State eigenvalue/eigenvector properties	Repeated theory component in 2015, 2018, 2023-2024.

Core preparation checklist

- Reduce a matrix to row echelon form, canonical/RREF form, then normal form; state rank at each stage.
- Find non-singular P and Q such that PAQ is in normal form.
- Find eigenvalues from $\det(A-\lambda I)=0$ and eigenvectors from $(A-\lambda I)X=0$.
- Know properties: eigenvalues of A^T , triangular matrices, A^{-1} , A^n ; largest eigenvalue variants.
- Practice diagonalization $P^{-1}AP = D$ when independent eigenvectors exist.

Secondary / rotating questions

- Cayley-Hamilton theorem may migrate into this set or Set 4.
- Algebraic/geometric multiplicity can appear as an extension.

Exam tactic

If you can reliably row-reduce and solve a $2 \times 2/3 \times 3$ characteristic polynomial, this is a very safe set. Prepare it as a compulsory Section A choice.

Vector Kinematics / Work Done + Cayley-Hamilton

A mixed set. Historically the “fourth slot” is more likely to rotate across mechanics/vector and theorem applications.

Historical footprint

Kinematics/space-curve questions appear in 2015, 2021, 2023, 2024, 2025-style papers. Cayley-Hamilton appears intermittently and is often paired with matrix inverse or polynomial evaluation.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Velocity/acceleration from a space curve	2015, 2021, 2023, 2024, 2025-style recurrence.
2	Tangent, normal, binormal, curvature/torsion, osculating plane	Strong in the dedicated vector-kinematics papers and current notes.
3	Work done along a path	Near-annual vector application across the decade.
4	Cayley-Hamilton verification / use	2018/2021/2025-type appearances; useful but less annual than rank/eigen blocks.
5	Force $\rightarrow$ acceleration/velocity	2015/2024/2025-style mechanics subquestion.

Core preparation checklist

- From $\mathbf{r}(t)$: compute $\mathbf{v}$, speed, $\mathbf{a}$, unit tangent $\mathbf{T}$, unit normal $\mathbf{N}$, binormal $\mathbf{B}$, curvature κ , radius of curvature, torsion τ and osculating plane.
- Use $\mathbf{F} = m\mathbf{a}$ to recover acceleration/velocity when initial data are given.
- Evaluate work $\int_C \mathbf{F} \cdot d\mathbf{r}$ along straight, piecewise or circular paths.
- State and verify Cayley-Hamilton; use it to compute $\mathbf{A}^{-1}$ or higher powers when asked.

Secondary / rotating questions

- Moment of force / vector triple product geometry.
- Direct coordinate-geometry of paths.

Exam tactic

Keep this as your Section A backup if you prefer the very stable matrix Sets 1-3. If you are strong in vectors, Set 4 can replace whichever matrix set feels weakest.

Risk note

This set has the highest topic-rotation risk in Section A; do not rely on only one subtopic inside it.

SECTION B

Recommended core: Set 5 is mandatory-level. Choose two from Sets 6, 7 and 8 based on your strength.

Two strong routes

Route L (Laplace-heavy): Sets 5 + 6 + 7. Route V (Vector-heavy): Sets 5 + 7 + 8. If beam/ODE applications are difficult, Route V is usually safer; if multiple integrals are difficult, Route L is usually safer.

Laplace Fundamentals & Direct Transform

The most reusable formula-and-property set in the paper.

Historical footprint

Laplace definition/direct-transform questions occur essentially every year, with periodic, piecewise and shifting variants recurring frequently. The “ $\sin(at)/t$ vs $\cos(at)/t$ ” existence question repeats notably in 2015 and 2024.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Direct Laplace transform using standard formulas/properties	Near-annual across 2015-2025.
2	Definition + sufficient conditions for existence	2017, 2018, 2019, 2022, 2025-type recurrence.
3	Periodic / piecewise Laplace transform	2019, 2021, 2022, 2024, 2025-style recurrence.
4	Unit step function / second shifting	2019, 2021, 2023, 2025 and topic-wise exam compilations.
5	First shifting / scaling / multiplication-division by t	Frequently asked as theory or embedded computation.
6	$L\{\sin at/t\}$; explain why $L\{\cos at/t\}$ does not exist	Direct repeat in 2015 and 2024; excellent high-priority theory item.

Core preparation checklist

- Know the definition $L\{f(t)\} = \int_0^\infty e^{-st}f(t)dt$ and sufficient existence conditions.
- Master standard transforms: 1 , t^n , e^{at} , $\sin at$, $\cos at$, $\sinh at$, $\cosh at$.
- Master linearity, first shifting, second shifting, scaling/change of scale, multiplication by t^n , division by t , transforms of derivatives/integrals.
- Practice periodic-function and piecewise-function transforms.
- Know unit-step representation and $L\{u(t-a)\} = e^{-as}/s$.

Secondary / rotating questions

- Graph-defined Laplace transforms.
- Transforms of derivatives and integrals.
- Gamma-function based transforms.

Exam tactic

Build a one-page formula sheet and practice recognizing which property is being tested. Set 5 is the most efficient Section B set to secure.

Inverse Laplace, Convolution & ODE / Beam Applications

High marks, highly repeated, but computationally longer.

Historical footprint

Inverse Laplace and convolution recur almost every year. Laplace-based ODE applications are also common; beam deflection appeared in 2017, 2018, 2019 and 2022 in the uploaded 2015-2024 beam compilation.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Inverse Laplace transform of rational functions	Near-annual across the paper set.
2	Convolution theorem / convolution-based inverse	Repeated in 2015, 2018, 2019, 2021-2025 variants.
3	Solve differential equation using Laplace transform	Recurring in 2016, 2017, 2019, 2020, 2021-2025.
4	Beam deflection using Laplace transform	2017, 2018, 2019, 2022; exactly four regular-exam appearances in the 2015-2024 beam compilation.
5	Heaviside expansion / shifted inverse	Common alternate inverse method.
6	Physical ODE application (cooling/building/forcing)	2025-style extension of the same transform workflow.

Core preparation checklist

- Inverse Laplace by partial fractions, completing the square, shifting and Heaviside expansion.
- State and apply convolution theorem: $L^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u)du$.
- Solve initial-value ODEs using transforms of y' , y'' , etc.
- For beam problems, start from EI $y^{(4)} = w(x)$, encode discontinuous loads with unit steps, then apply boundary conditions.
- Know boundary conditions: clamped $y=y'=0$; hinged $y=y''=0$; free $y''=y'''=0$.

Secondary / rotating questions

- Integral equations solved by Laplace.
- Special inverse forms involving logarithms.

Exam tactic

If you are comfortable with algebraic partial fractions, this set is excellent. If beam/ODE work is a weakness, treat Set 6 as the backup and use Sets 5-7-8 instead.

Risk note

Long algebra is the main risk. Show transform setup and boundary/initial conditions clearly for method marks.

Vector Differential Calculus

Gradient, directional derivative, surfaces and conservative fields form one of the most persistent vector blocks.

Historical footprint

Directional derivatives, tangent/normal geometry, solenoidal/irrotational tests and scalar potential questions recur throughout the decade, including the most recent papers.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Directional derivative / maximum rate of change	Repeated 2016, 2017, 2019, 2020, 2022, 2023, 2025-style.
2	Angle between surfaces / orthogonal surfaces	Repeated in 2015, 2016, 2018, 2021, 2022, 2023.
3	Solenoidal / irrotational / conservative + potential	Repeated across 2015, 2017-2019, 2021-2023, 2025.
4	Tangent plane / normal line	Repeated 2016, 2017, 2021-2022 and related variants.
5	Vector identities involving div/curl/grad	Theory/proof family appearing in multiple papers.
6	Physical meaning of $\nabla\phi$ and $\nabla\cdot\mathbf{F}$	Explicit 2015 theory; still important short-answer material.

Core preparation checklist

- Compute $\nabla\phi$; identify maximum directional derivative and its direction.
- Find directional derivative along a given vector after normalizing it.
- Use gradients to find tangent/normal planes and angle/orthogonality of two surfaces.
- Test a field for solenoidal ($\nabla\cdot\mathbf{F}=0$) and irrotational ($\nabla\times\mathbf{F}=0$).
- If irrotational, find scalar potential ϕ with $\mathbf{F}=\nabla\phi$.
- Know physical meaning of gradient, divergence and curl.

Secondary / rotating questions

- Laplacian $\nabla^2\phi$ / ∇^2f .
- Moment of force and triple product geometry.

Exam tactic

This is a strong Section B anchor because the methods are short and standardized: differentiate, substitute the point, normalize the direction, and take dot/cross products.

Vector Integration: Line, Surface & Volume

The major integration set. Historically worth large marks and highly patterned.

Historical footprint

The uploaded 2015-2024 vector-integration compilation identifies line/work, surface flux and volume integrals as a core exam component, typically accounting for roughly 28-40 marks in a term-final paper.

Most repeated / highest-return question families

Rank	Question family	Observed pattern / years
1	Work done / line integral along straight or piecewise paths	Appears repeatedly from 2015 onward; one of the most durable vector applications.
2	Surface flux over plane/sphere/cylinder, often first octant	Repeated 2015, 2018, 2019, 2021-2025 variants.
3	Volume integral over bounded 3D region	Repeated 2015, 2019, 2021-2024 and compiled as a core topic.
4	Line integral on spiral/circle/triangle paths	Recurring geometry variants; current notes contain several.
5	Closed-surface / divergence-theorem style flux	A natural recurring extension of surface integration.
6	Area/region setup from intersecting planes/cylinders	Common source of marks and common source of mistakes.

Core preparation checklist

- Line integrals/work: parameterize C , compute $d\mathbf{r}$, substitute $F(\mathbf{r}(t))$, then integrate $F \cdot d\mathbf{r}$.
- Piecewise straight paths: split C into segments and evaluate separately.
- Surface flux: choose a convenient projection, compute unit normal or vector area element, set correct bounds.
- Volume integrals: sketch the region, choose order of integration, derive limits carefully.
- Know when Gauss divergence / Stokes theorem can simplify a closed surface or boundary integral.

Secondary / rotating questions

- Stokes theorem verification.
- Surface-area integral vs flux integral distinctions.

Exam tactic

Write the parameterization and limits before doing algebra. Correct setup earns substantial method marks even if the final integral becomes long.

Master Ranking: Most Repeated Question Families

#	Question family	Area	Frequency	Action
1	Consistency / unique-no-infinite solution using rank or a parameter	Matrix	Near-annual	Must prepare
2	Eigenvalues + eigenvectors / properties / largest eigenvalue	Matrix	Near-annual	Must prepare
3	Direct Laplace transform using standard properties	Laplace	Near-annual	Must prepare
4	Inverse Laplace transform	Laplace	Near-annual	Must prepare
5	Work done / line integral along a curve	Vector	Near-annual	Must prepare
6	Matrix definitions with examples	Matrix	Very frequent	Must prepare
7	Echelon / canonical / normal form + rank	Matrix	Very frequent	Must prepare
8	Directional derivative / gradient / maximum rate	Vector	Very frequent	Must prepare
9	Solenoidal-irrotational-conservative field / scalar potential	Vector	Very frequent	Must prepare
10	Surface integral / flux	Vector	Very frequent	Must prepare
11	Convolution theorem / convolution-based inverse	Laplace	Frequent	High priority
12	Inverse matrix by elementary row transformations	Matrix	Frequent	High priority
13	Laplace ODE / IVP	Laplace	Frequent	High priority
14	Angle between surfaces / tangent-normal geometry	Vector	Frequent	High priority
15	Periodic / piecewise / unit-step Laplace	Laplace	Frequent	High priority

Targeted recurring application

Beam deflection via Laplace appeared in four regular exams in the uploaded 2015-2024 beam compilation (2017, 2018, 2019 and 2022). It is not annual, but when it appears it carries high marks.

Additional: Theory Question List

Why theory matters

The uploaded 2015-2024 theory compilation estimates roughly 30-40 marks per exam can come from definitions, properties, theorem statements or proofs. These are efficient marks and should be prepared as short, exam-ready answers.

Matrix / Linear Algebra

- Definitions with example: equivalent, singular/non-singular, trace, symmetric, Hermitian, scalar, orthogonal, idempotent, involutory, nilpotent, skew-symmetric, elementary, diagonal, invertible.
- Rank of a matrix; echelon / canonical / normal form.
- Consistent vs inconsistent system; trivial vs non-trivial solution.
- Linear dependence / independence.
- Eigenvalue and eigenvector definitions; eigenvalue properties.
- Elementary row/column transformations; equivalent matrices.

Vector Calculus

- Physical meaning of gradient, divergence and curl.
- Solenoidal, irrotational and conservative vector fields.
- Scalar triple product: geometrical meaning.
- Unit tangent, normal, binormal; curvature, torsion, osculating plane.
- Directional derivative and maximum directional derivative.
- Tangent plane and normal line to a surface.

Laplace Transform

- Definition of Laplace transform and kernel of the transform.
- Sufficient conditions for existence: piecewise continuity + exponential order.
- Linearity, first shifting, second shifting, scaling, multiplication by t^n , division by t .
- Transform of derivatives and integrals.
- Unit step / Heaviside function and its transform.
- Convolution theorem and inverse-transform form.
- Periodic function transform.
- Boundary-condition meanings for beam deflection.

Additional: Proof / Derivation Priority List

Priority	Proof / derivation	Evidence / use
★★★★★	A square matrix can be expressed uniquely as symmetric + skew-symmetric	2017, 2018; concept returns in later papers
★★★★★	If A is symmetric and non-singular, then A^{-1} is symmetric	2016; classic short proof
★★★★★	Diagonal elements of a skew-symmetric matrix are zero	2018; foundational proof
★★★★☆	If A is involutory, $(I+A)/2$ and $(I-A)/2$ are idempotent	2018
★★★★☆	Direction-cosine matrix of three mutually perpendicular lines is orthogonal	2022
★★★★☆	Vector triple product identity and Jacobi identity	2016
★★★★☆	$\text{div}(\phi A) = A \cdot \text{grad } \phi + \phi \text{ div } A$	Recurring vector-identity style
★★★★☆	$\text{curl}(\text{grad } \phi) = 0$ / conservative field implications	Core potential-field derivation
★★★★☆	Convolution theorem statement and application	Repeated theorem/application family
★★★★☆	Why $L\{\cos(at)/t\}$ does not exist; $L\{\sin(at)/t\}$ exists	2015 and 2024 direct repeat
★★★★☆☆	Use Laplace concepts to evaluate standard definite integrals (e.g. $\sin^2 t/t^2$)	2019-style
★★★★☆☆	Cayley-Hamilton theorem verification and use for inverse/powers	Intermittent but high-value
★★★★☆☆	Beam deflection transform derivation from EI $y^{(4)} = w(x)$	2017/2018/2019/2022 application family

3-out-of-4 Answer Strategy

Rule

Your goal is not to master all four sets equally. Build three “exam-ready” sets per section and one backup set. A set is exam-ready only when you can solve both its theory and computational variants under time pressure.

Section A - default route

- 1. Set 1: Matrix Fundamentals & Core Theorems
- 2. Set 2: Inverse, Rank, Consistency & Dependence
- 3. Set 3: Normal Form & Eigenvalue Block
- Backup: Set 4 if a Set 1-3 question looks unusually long or unfamiliar.
- Reason: Sets 1-3 are historically the most stable matrix clusters and share row-reduction skills.

Section B - choose your profile

Profile	Prepare	Skip/backup	Best for
Laplace-strong	5 + 6 + 7	8	Fast algebra, ODEs, transforms
Vector-strong	5 + 7 + 8	6	Geometry, vector calculus, multiple integrals

Time allocation inside the exam

- Start with the set whose entire solution path is obvious.
- Do short theory/definition subparts first to lock marks quickly.
- For matrix row-reduction and vector integration, write the setup clearly before arithmetic.
- If one subpart stalls for several minutes, move to the next subpart; the 3-of-4 structure rewards breadth and selection.
- Keep one prepared backup set in each section so you never become dependent on a single difficult variant.

Year-wise Trend Notes (2015-2025)

Section labels are not fixed topic labels The papers demonstrate that Matrix, Vector and Laplace blocks move between Section A and Section B. The safest preparation is therefore topic-set based, not “Section A always equals Matrix” or “Section B always equals Laplace”.		
Year	Dominant section pattern	Takeaway
2015	A heavily Matrix; B Vector + Laplace	Classic 4+4 structure
2016	A Matrix; B Vector + Laplace	Strong baseline year
2017	A Laplace/Vector; B Matrix	Clear section swap
2018	A Matrix; B Laplace/Vector	Matrix-heavy A returns
2019	Mixed A; Vector/application-heavy B	Beam + Laplace + vector spread
2020	Shortened 120-mark paper	Special 3+3 structure; use only as supporting pattern
2021	A Vector/Matrix; B Laplace/Matrix	Strong mixing
2022	A Vector + Matrix; B Laplace + applications	Beam reappears
2023	A Vector/Laplace; B Matrix/Kinematics	Very clear modern distribution
2024	A Vector/Laplace; B Matrix/Kinematics	Very similar to 2023
2025	A Matrix + Kinematics; B Laplace + Vector integration	Latest switch; basis of this 8-set map

Final 7-Day Revision Checklist

Day	Focus	Minimum output
Day 1	Set 1 + theory definitions	Write every matrix definition from memory; do 2 decomposition/proof problems.
Day 2	Set 2	Do 2 inverse-by-ERO + 3 consistency/rank problems.
Day 3	Set 3	Do 2 normal-form/rank + 3 eigenvalue/eigenvector problems.
Day 4	Set 5	Memorize transform table/properties; do periodic + unit-step + direct transform.
Day 5	Set 7	Directional derivative, surfaces, potential field, vector identities.
Day 6	Choose Set 6 or Set 8 as third B-set	Either ILT/convolution/ODE/beam OR line/surface/volume integrals.
Day 7	Backups Set 4 + remaining B-set	One timed mock: choose 3/4 in each section before solving.

Last-minute non-negotiables

- Matrix: rank-consistency theorem, inverse by ERO, normal form, eigenvalue workflow.
- Laplace: standard table, shifting, derivative formulas, convolution, unit step.
- Vector: grad/div/curl, directional derivative, potential, line/surface integral setup.
- Beam: end boundary conditions and EI $y^{(4)}=w(x)$.
- Theory: definitions must include the defining equation/condition and one compact example.

Source Basis & Method Note

- Primary: "Math 2123 Term Questions.pdf" (KUET regular exam papers, 2015-2025).
- Supporting frequency/solution collections: "kuet-math3-theory-and-proofs.pdf", "kuet-math3-vector-integrations.pdf", "kuet-math3-beam-deflections.pdf", "kuet-math3-beam-deflections-udl-uvl.pdf", and "kuet-math3-unit-step-functions.pdf".
- Lecture-note validation: uploaded Matrix, Vector, Laplace, inverse Laplace and teacher/student note sets.
- Reference validation: Murray R. Spiegel (Vector Analysis; Laplace Transforms), H. K. Dass, Shanti Narayan, Frank Ayres Jr., David C. Lay and P. N. Chatterjee uploads.

Interpretation caution

"Most repeated" means repeated question family / method, not always word-for-word repetition. Exact matrices, constants, surfaces and functions change. Section placement also changes across years. Prepare the method, not only a memorized numerical answer.

QUESTION BANK ANALYSIS

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Prepared for strategic KUET Math 2123 term-final revision